EduAssist AI — LLMOps: API Cost, Latency, and Prompt Version Management

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Course	AIMLCZG523 — MLOps
Assignment	02 (LLMOps)
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Git Repo	https://github.com/2025ae05369-badree/eduassist-llmops-metrics.git
Google drive link	https://wilpbitspilaniacin0-my.sharepoint.com/:f:/g/personal/2025ae05369_wilp_bits-pilani_ac_in/IgCxosIq1svlQYqMfcZ_PDCzAYx1izEtKQIY5gx4BBetz2o

1. LLMOps vs MLOps

MLOps manages datasets, feature pipelines, training, deployment and monitoring of models the team trains itself. LLMOps extends this to applications built on large pre-trained models, where the artefacts that change most are not weights but *prompts*, model/API versions, retrieved context and output guardrails. It adds per-token economics (input and output tokens are billed separately), latency driven by generation length, and non-deterministic, free-form outputs that need automated checks such as JSON validation. Prompts therefore must be versioned, evaluated and promoted like models.

2. Setup

A paid hosted API needs an account with billing, which was unavailable, so — as permitted by §7 of the brief — the pipeline uses the open-weight, ungated `Qwen/Qwen2.5-3B-Instruct` (revision `aa8e7253`) running locally through Hugging Face `transformers`. The "API version" is recorded as `transformers 5.22.1`, `torch 2.15.0`, `mlflow 3.17.2` (Python 3.12.6), on an Apple M1 Pro (MPS, float16), with greedy decoding and `max_new_tokens = 512`. A pilot with the 1.5B variant wrapped every JSON answer in code fences, so the 3B model was chosen. Tokens are counted with the model's tokenizer. Because no per-token price is published for this model, prices were derived from the Hugging Face Inference Endpoints rate for one NVIDIA T4 (**$0.50/hour, USD, pricing date 2026-11-02**) and the measured prefill (589.9 tokens/s) and decode (22.4 tokens/s) throughput: **$0.2354 per 1M input** and **$6.2116 per 1M output** tokens.

3. Latency findings (Task 1, prompt v1)

After an untimed warm-up, five requests averaged **18.44 s** (median 22.58 s, min 8.05 s, max 22.89 s, std 6.48 s). The slowest was Q1 (overfitting, 22.89 s, 512 output tokens); the fastest was Q2 (Newton's second law, 8.05 s, 166 tokens). Input lengths were nearly constant (46–55 tokens), and latency correlated almost perfectly with output tokens (r = 0.999) at ~22 tokens/s. Three answers hit the 512-token cap and were truncated.

4. Token and cost findings (Task 2)

The average v1 request used 51.2 input and 413.8 output tokens, costing **$0.002582**; at 10,000 requests per month the projected cost is **$25.82**. Output tokens were 89.0% of tokens but 99.5% of cost. The recommended optimisation is therefore to **shorten responses** — concise structured prompts plus a lower token cap — rather than trimming context or caching the prompt prefix, which

together account for only 0.5% of cost. Routing simple factual questions to a smaller model would reduce it further.

5. Prompt versioning results (Task 3)

Three versions of `eduassist-qa` were registered in the MLflow Prompt Registry with commit messages, evaluated in experiment `eduassist_llmops_assignment` on the same five questions, and logged with the required metrics and per-question response artefacts.

Version	Avg latency (s)	Avg input tokens	Avg output tokens	Avg cost (USD)	valid_json	Recommended
v1 Basic	18.87	51.2	413.8	0.002582	0.0	No
v2 Structured	12.06	73.2	270.2	0.001696	0.0	No
v3 JSON	5.64	107.2	123.4	0.000792	1.0	Yes

6. Promotion decision

Version 3 was promoted to the `production` alias using a rule of JSON validity first, then cost, then latency. It is the only version whose five responses parsed as schema-conformant JSON, and its fixed structure also makes it the fastest (3.3× faster than v1) and cheapest (≈ $7.92 per 10,000 requests, −69% vs v1), despite ~56 extra input tokens. Its weaknesses are an omitted complexity analysis for Q3, keyword-style key points and uniformly "high" confidence; a future v4 should address these and be promoted only if it beats v3 on the same evaluation.

7. Limitations

- **Non-determinism:** greedy decoding fixes token outputs, but latencies vary between runs; hosted APIs add sampling and server-side variability.
- **Sample size:** five questions give indicative, not statistically robust, comparisons.
- **Hardware:** single-stream laptop latency excludes network and queueing and is not representative of a hosted, batched API.
- **Estimated cost:** prices are derived from an hourly GPU rate and local throughput, not billed; batched serving would be much cheaper per token.
- **Small-model JSON reliability:** compliance depended on model size (1.5B failed); production still needs validation and retries.